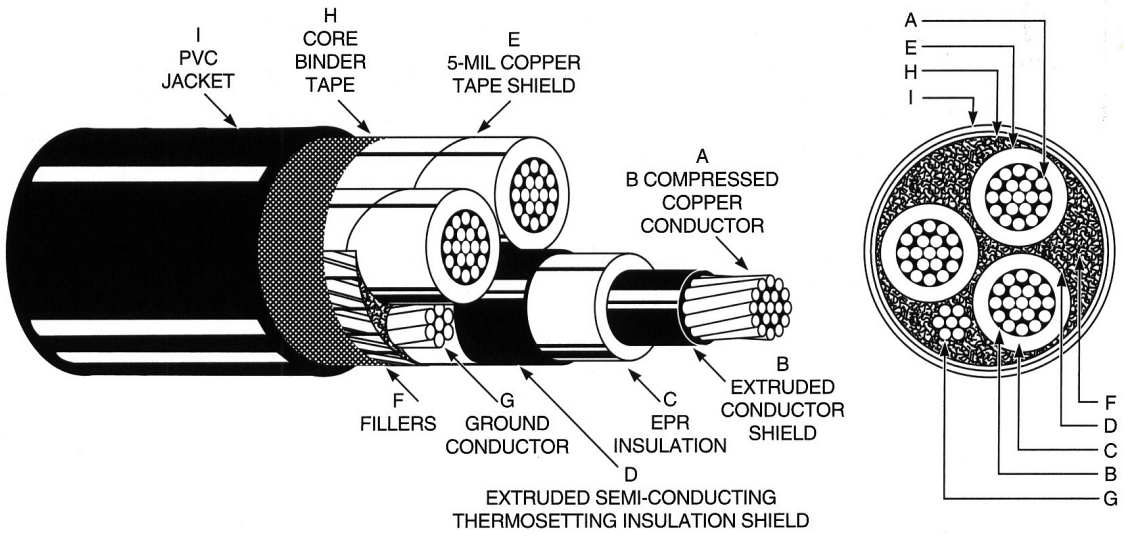
CT3-09ET

**5kVU or 8kV Type MV-105. 3 Copper Conductors.
Thermosetting Conductor and Insulation Shield.
EPR insulation. Tape Shield.
Copper Ground Wire. PVC Jacket.
RoHS**



APPLICATIONS

Southwire CT3-09ET Type MV-105 Cable is for use in aerial installations, direct buried, cable trays, troughs or raceway. These cables are capable of operating continuously at a maximum conductor temperature of 105°C for normal operation, 140°C for emergency overload conditions, and 250°C for short circuit conditions, and are rated at 5,000V, 133% (grounded system) and 8,000V, 100% insulation levels (grounded system). Maximum sidewall pressure is 1000 lbs.

SPECIFICATIONS

Southwire CT3-09ET Type MV-105 Cable is manufactured and tested in accordance with the latest revisions of the following standards and specification:

- UL 1072 and 1685
- ICEA S-93-639 (NEMA WC 74)
- ICEA S-97-682 (when requested)
- IEEE 1202/FT4 - (Sizes 4 AWG and larger)
- AEIC CS-8
- RoHS

CONSTRUCTION

Southwire CT3-09ET Type MV-105 Cable offers heavy duty construction and flexible EPR insulation. For use in cable trays per NEC Article 392. Alternate constructions available. SOLONON low smoke, non halogen polyolefin jackets and CPE jackets are available upon request.

Southwire
One Southwire Drive
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CT3-09ET 3c, 5/8kV, 115mil EPR(133/100%), TS, GW, PVC, CT

Product Code	Size (AWG or kcmil)	Diameter		0.115" (2.92 mm) Insulation Diameter		Extruded Insulation Shield Diameter		Grd. Cond. Size	Min. Point Jacket Thickness		Approximate Overall Diameter		Approximate Net Weight		Allowable Ampacities+	
		in	mm	in	mm	in	mm	AWG	in	mm	in	mm	lb./Mft.	kg/km	Duct	Conduit In Air
CT3-09ET-006	6	.178	4.52	.465	11.81	.52	13.21	6	.07	1.78	1.329	33.8	1007	1498	95	92
CT3-09ET-004	4	.225	5.72	.513	13.02	.568	14.41	6	.07	1.78	1.427	36.2	1234	1836	125	120
CT3-09ET-001	1	.322	8.18	.608	15.43	.663	16.83	4	.07	1.78	1.637	41.6	1836	2732	185	185
CT3-09ET-030	3/0	.456	11.58	.738	18.73	.793	20.13	3	.1	2.54	1.981	50.3	3006	4473	270	280
CT3-09ET-040	4/0	.512	13	.793	20.13	.848	21.53	3	.1	2.54	2.100	53.3	3516	5232	305	320
CT3-09ET-250	250	.558	14.17	.85	21.59	.905	22.99	3	.07	1.78	2.236	56.8	4055	6033	335	350
CT3-09ET-350	350	.661	16.79	.953	24.19	1.008	25.59	3	.07	1.78	2.457	62.4	5202	7740	400	430
CT3-09ET-500	500	.79	20.07	1.078	27.37	1.133	28.77	3	.07	1.78	2.780	70.6	7058	10502	485	525
CT3-09ET-750	750	.968	24.59	1.265	32.13	1.32	33.53	3	.07	1.78	3.185	80.9	9905	14738	585	635

*Nominal diameter per ASTM Standards. ***±0.030" ***±0.050"
+Ampacities are based on the NEC, 2008 Edition. Duct ampacities are based on Table 310.79 three conductors within an overall covering in one underground duct,
105°C conductor, 20°C earth ambient temperature. Conduit in air ampacities are based on Table 310.75 three cables within an overall covering in isolated conduit in
air, 105°C conductor, 40°C ambient temperature.

Scope: This specification covers three conductor EPR (ethylene propylene rubber) insulated, shielded, thermoplastic jacketed power cables with grounding conductor for use in aerial installations, direct buried, cable trays, troughs, or raceways. These cables are capable of operating continuously at a temperature of 105°C for normal operations, 140°C for emergency overload conditions, and 250°C for short circuit conditions, and are rated at 8,000V, 100% (grounded system) and 5000V, 133% insulation levels (ungrounded system). ,

Standards: The following standards shall form part of this specification - UL Standard 1072 for Medium Voltage Power Cable and ICEA S-93-639 (NEMA WC 74) 5-46 kV Shielded Power Cable for Use in the Transmission & Distribution of Electric Energy.

Conductor: The conductor shall be Class B compressed concentric stranded bare copper in accordance with ASTM B3 and B8 and ICEA Part 2, Section 2.1 and 2.5.

Conductor Shield: The conductor shall be shielded with an extruded semi-conducting thermosetting polymeric layer, which shall be firmly bonded to the insulation. The thickness shall be in accordance with the referenced standards.

Insulation: The insulation shall be EPR (ethylene propylene rubber) meeting the requirements of the referenced standards. The nominal thickness shall be 0.115".

Insulation Shield: The insulation shall be shielded with an extruded semi-conducting thermosetting polymeric layer which shall be identified as semi-conducting. Over this layer shall be applied a helically wrapped 5-mil copper tape with 25% overlap. The method of phase identification will be similar to ICEA Method 3, using printed circuit numbers and colors (1-BLACK, 2-RED, 3-BLUE).

Grounding Conductor: The ground conductor shall be Class B compressed concentric stranded bare copper in accordance with ASTM B3 and B8.

Assembly: The insulated and shielded power conductors shall be cabled round with fillers and with a grounding conductor in one outer interstice and covered with a binder tape.

Jacket: The cable shall be covered with a black PVC jacket conforming to the requirements specified for polyvinyl chloride in ICEA. The average thickness shall be in accordance with the referenced standards. The jacket will be sunlight resistant and will meet the requirements of UL 1685 - UL Flame Exposure Test. Optional SOLONON low smoke, non halogen polyolefin jackets and CPE jackets are available upon request.

Tests: Qualification tests shall be conducted in accordance with the requirements of AEIC.

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